

REGULAR EXPRESSIONS (REGEX)

Chapter 1: Introduction to Regular Expressions

What is a Regular Expression?

A Regular Expression (Regex) is a sequence of characters that defines a search pattern.

It is used for:

- Searching text
- Matching patterns
- Validating input
- Extracting information
- Replacing text

Python provides the **re** module to work with regular expressions.

```
import re
```

Chapter 2: Match Function

Theory

The `re.match()` function checks whether a pattern exists at the beginning of a string.

Syntax

```
re.match(pattern, string)
```

Returns

- Match Object → If pattern found at beginning
 - None → If not found
-

Example 1

```
import re
```

```
text = "Python Programming"
```

```
result = re.match("Python", text)
```

```
if result:
```

```
    print("Match Found")
```

```
else:
```

```
    print("No Match")
```

Output

```
Match Found
```

Example 2

```
import re
```

```
text = "I love Python"
```

```
result = re.match("Python", text)
```

```
print(result)
```

Output

None

Chapter 3: Search Function**Theory**

The `re.search()` function searches the entire string and returns the first match.

Syntax

```
re.search(pattern, string)
```

Example 1

```
import re
```

```
text = "I love Python"
```

```
result = re.search("Python", text)
```

```
if result:
```

```
    print("Found")
```

```
    print("Position:", result.start())
```

Output

Found

Position: 7

Example 2 – Find Email

```
import re
```

```
text = "Contact us at info@gmail.com"
```

```
email = re.search(r'\w+@\w+\.\w+', text)
```

```
print(email.group())
```

Output

info@gmail.com

Match vs Search

Feature	match()	search()
Checks Beginning	Yes	No
Searches Entire String	No	Yes
Returns Match Object	Yes	Yes

Example

```
import re
```

```
text = "I love Python"
```

```
print(re.match("Python", text))  
print(re.search("Python", text))
```

Output:

None

<Match Object>

Chapter 4: Grouping**Theory**

Parentheses () create groups.

Groups help extract specific parts of a match.

Syntax

(pattern)

Example 1 – Extract Date

```
import re
```

```
text = "15-07-2025"
```

```
pattern = r'(\d{2})-(\d{2})-(\d{4})'
```

```
match = re.search(pattern, text)
```

```
print("Day:", match.group(1))
```

```
print("Month:", match.group(2))
```

```
print("Year:", match.group(3))
```

Output

Day: 15

Month: 07

Year: 2025

Example 2 – Extract Name and Age

```
import re
```

```
text = "John 25"
```

```
match = re.search(r'([A-Za-z]+)\s(\d+)', text)
```

```
print("Name:", match.group(1))
print("Age:", match.group(2))
```

Output

Name: John

Age: 25

Chapter 5: Matching at Beginning or End**Theory**

Regex provides special symbols:

Symbol	Meaning
^	Start of String
\$	End of String

Beginning Match (^)

```
import re
```

```
text = "Python Programming"
```

```
if re.search(r'^Python', text):
    print("Starts with Python")
```

Output:

Starts with Python

End Match (\$)

```
import re
```

```
text = "Learn Python"
```

```
if re.search(r'Python$', text):
    print("Ends with Python")
```

Output:

Ends with Python

Username Validation

```
import re
```

```
username = "Athar123"
```

```
pattern = r'^[A-Za-z0-9]+$'
```

```
if re.match(pattern, username):
    print("Valid Username")
```

else:

```
print("Invalid Username")
```

Output:

Valid Username

Chapter 6: Match Objects

Theory

Regex functions return a Match Object.

Useful methods:

Method	Purpose
group()	Matched text
start()	Start index
end()	End index
span()	Start and End

Example

```
import re
```

```
text = "Welcome to Python"
```

```
match = re.search("Python", text)
```

```
print("Matched:", match.group())
```

```
print("Start:", match.start())
```

```
print("End:", match.end())
```

```
print("Span:", match.span())
```

Output

Matched: Python

Start: 11

End: 17

Span: (11, 17)

Chapter 7: Flags

Theory

Flags modify regex behavior.

Common Flags

Flag	Meaning
re.I	Ignore Case
re.M	Multiline
re.S	Dot Matches Newline
re.X	Verbose Mode

Ignore Case (re.I)

```
import re
```

```
text = "python"
```

```
result = re.search("Python", text, re.I)
```

```
print(result.group())
```

Output:

```
python
```

Multiline (re.M)

```
import re
```

```
text = """Python
```

```
Java
```

```
C++"""
```

```
result = re.findall(r'^Java', text, re.M)
```

```
print(result)
```

Output:

```
['Java']
```

Dotall (re.S)

```
import re
```

```
text = "Hello\nWorld"
```

```
result = re.search(r'Hello.*World', text, re.S)
```

```
print(result.group())
```

Output:

```
Hello
```

```
World
```

Common Regex Symbols

Symbol	Meaning	Example
.	Any character	a.b
\d	Digit	\d+
\D	Non-digit	\D+
\w	Word character	\w+
\W	Non-word character	\W+
\s	Whitespace	\s
\S	Non-whitespace	\S

+	One or More	a+
*	Zero or More	a*
?	Optional	colou?r
{n}	Exactly n times	\d{4}
{n,m}	Between n and m	\d{2,5}
[]	Character Set	[abc]
()	Grouping	(abc)
^	Start	^Python
\$	End	Python\$

Real-World Applications

1. Email Validation

```
import re

email = "student@gmail.com"

pattern = r'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$'

if re.match(pattern, email):
    print("Valid Email")
else:
    print("Invalid Email")
```

2. Phone Number Validation

```
import re

phone = "9876543210"

pattern = r'^[6-9]\d{9}$'

if re.match(pattern, phone):
    print("Valid Number")
else:
    print("Invalid Number")
```

3. Extract URLs

```
import re

text = """
Visit https://google.com
Visit https://github.com
Visit https://openai.com
"""
```

```
urls = re.findall(r'https?://[^\s]+', text)
```

```
for url in urls:
```

```
    print(url)
```

Output

```
https://google.com
```

```
https://github.com
```

```
https://openai.com
```

Mini Project: Extract Information from Text

```
import re
```

```
text = """
```

```
Name: Athar Ahmed
```

```
Email: athar@gmail.com
```

```
Phone: 9876543210
```

```
"""
```

```
name = re.search(r'Name:\s([A-Za-z ]+)', text)
```

```
email = re.search(r'\w+@\w+\.\w+', text)
```

```
phone = re.search(r'\d{10}', text)
```

```
print("Name :", name.group(1))
```

```
print("Email:", email.group())
```

```
print("Phone:", phone.group())
```

Output

```
Name : Athar Ahmed
```

```
Email: athar@gmail.com
```

```
Phone: 9876543210
```

Interview Questions

Q1. What is Regex?

A Regular Expression is a sequence of characters used for pattern matching and text processing.

Q2. Difference between match() and search()?

- match() checks only the beginning.
- search() checks the entire string.

Q3. What is a Match Object?

An object returned after successful matching containing information about the match.

Q4. What are Groups?

Parentheses () used to capture parts of a matched pattern.

Q5. What does ^ mean?

Matches the beginning of a string.

Q6. What does \$ mean?

Matches the end of a string.

Q7. What is re.I?

Ignore Case flag.

Q8. What is re.M?

Multiline flag.

Q9. What is re.S?

Dotall flag.

Q10. How do you extract emails using Regex?

```
re.findall(r'\w+@\w+\.\w+', text)
```

One-Page Revision Sheet

Topic	Function/Symbol
Match	re.match()
Search	re.search()
Find All	re.findall()
Replace	re.sub()
Split	re.split()
Grouping	()
Start Match	^
End Match	\$
Digit	\d
Word	\w
Space	\s
Ignore Case	re.I
Multiline	re.M
Dotall	re.S